

LSGI3262 Laser Scanning Lab 2

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Task 1

Using ArcGIS Pro, select three land-cover classes from the airborne LiDAR intensity datasets at 532 nm, 1064 nm, and 1550 nm. Please show the screen shots of the polygons you selected for the land cover polygons. (2 points)



Figure 1: The polygons of the three land cover classes selected. From left to right: SoilCover, GrassCover, TreeCover.

Task 2

Plot a bar chart of the three land cover objects by showing mean, min, and max, and comment on their intensity values with respect to different wavelengths. (2 points)

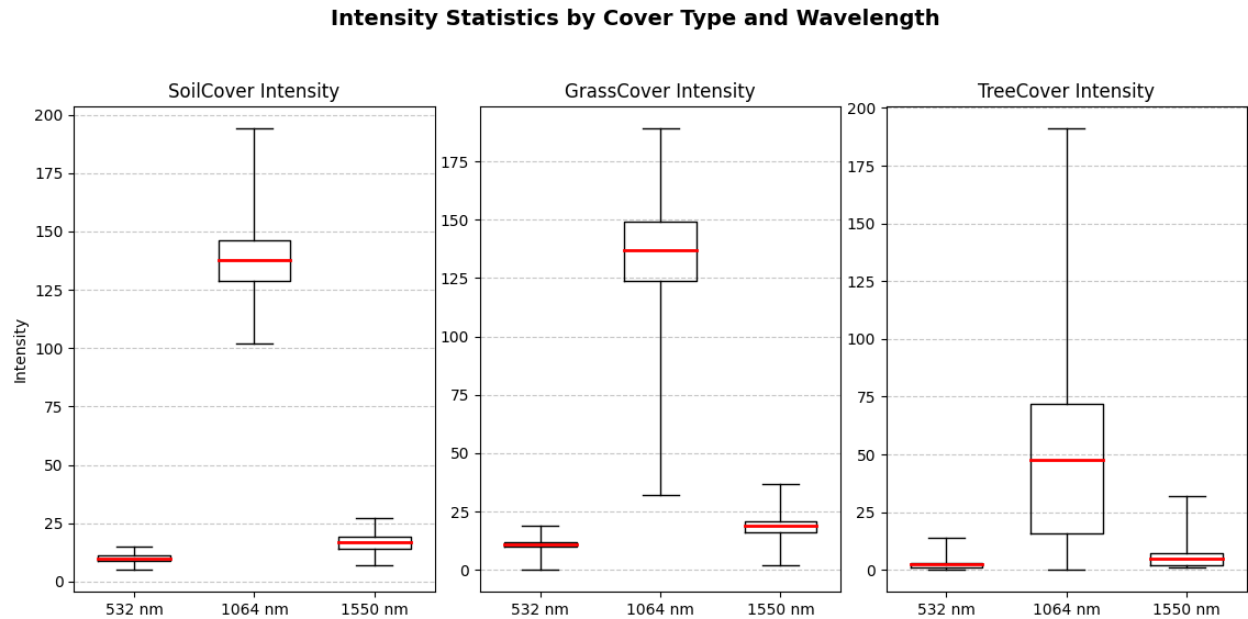


Figure 2: showing the mean, min, and max intensity values of the 3 land cover objects. The figure also shows first and third quartiles of the intensity values to plot a box and whisker diagram.

It appears that 1064 nm wavelength has the highest intensity and variation across all materials, while 532 nm and 1550 nm wavelengths have lower intensity and the data are more clustered. This is likely due to high reflectance for 1064nm wavelength for most materials. 532nm has the lowest in intensity, this is likely due to the low reflectance in the spectrum and concern of eye safety.

Task 3

Retrieve the corresponding spectral reflectance of these three land cover objects from a spectral library and perform a correlation with respect to the wavelength. (2 points)

Since only the intensity of points is provided, the spectral reflectance can be approximated with the following formula:

Since there are outliers in the dataset, the maximum intensity is set to the maximum of 3σ range. The intensity is the mean intensity in the polygons.

Statistics of LiDAR intensity datasets

Wavelength	Mean	Standard deviation	3σ range
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532nm	4.40	4.22	[0.0, 17.06]
1064nm	61.84	48.11	[0.0, 206.17]
1550nm	6.63	15.97	[0.0, 54.54]

By applying the formula, we have the approximated spectral reflectance in the following table.

Land cover polygons	Spectral reflectance at wavelength		
	532nm	1064nm	1550nm
SoilCover	0.58	0.67	0.31
GrassCover	0.64	0.66	0.34
TreeCover	0.14	0.23	0.09

The above table combined with the actual spectral reflectance of the materials from Jet Propulsion Laboratory of NASA, the correlation figure can be plotted as below.

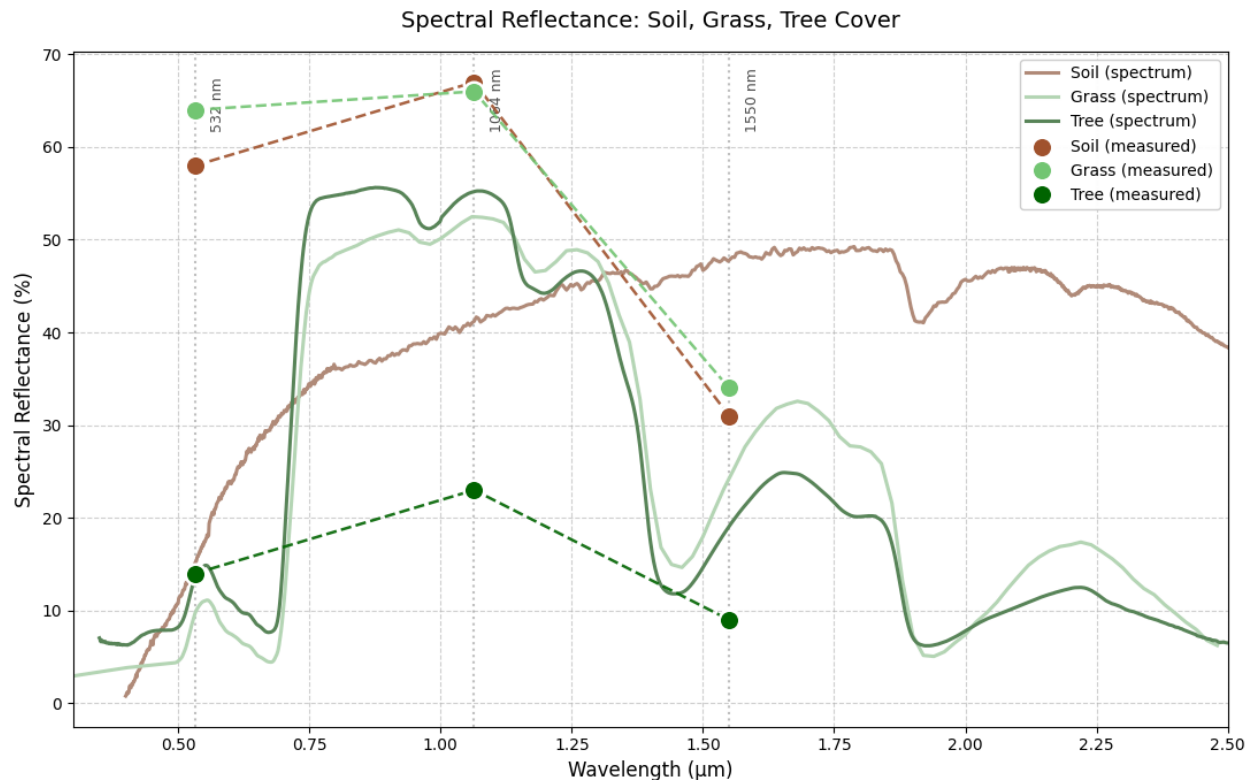


Figure 3: Spectral reflectance from NASA statistics and approximations, by land cover material and wavelength.

From the figure, it appears that all 3 materials changes the spectral reflectance from low to high, then to low to some extent, as the wavelength increases from 0.5 to 1.5 microns. For grass and tree, the pattern matches the expected low-high-low pattern, with moderate to high deviations in values. The trees are likely Coniferous trees, since the reflectivity is typically lower than

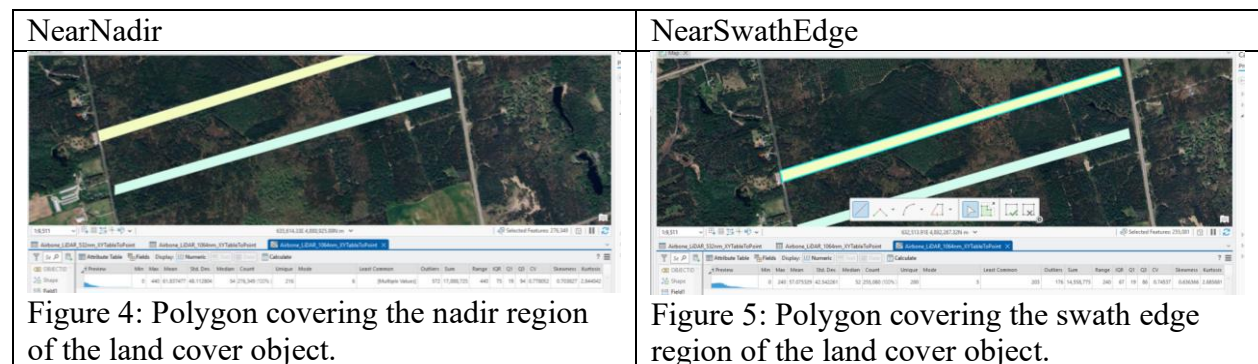
Deciduous trees, around 30% at NIR. However, for approximated spectral reflectance of soil also exhibits a low-high-low pattern from 0.5 to 1.5 microns, different from the expected increasing pattern.

In a nutshell, the figure shows that there is low to moderate correlation between the spectral library and the approximated values.

Task 4

Select any one of the land cover objects, then draw two respective polygons on the lidar intensity data on the close-to-nadir region and swath edge. Retrieve their lidar intensity statistics, and comment on it (e.g., which region has higher lidar intensity even though they are on the same land cover object, and why?). (4 points)

A woodland was selected as the land cover object. Below shows the screenshots of the polygons of close-to-nadir region and swath edge, and their LiDAR intensity statistics.



LiDAR intensity statistics

	Max	Min	Mean	Q1	Q3
Nadir	440	0	61.83	19	94
Swath Edge	240	0	57.08	19	86

Intensity Statistics Nadir vs Swath Edge

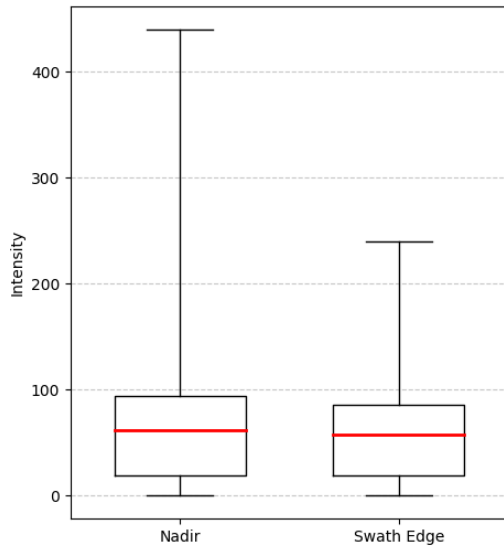


Figure 6: LiDAR intensity statistics, showing the max, min, and mean intensity values.

From the statistics, the nadir region has a higher mean intensity (61.83) compared to the swath edge (57.08). The maximum intensity is also higher at nadir (440) than at swath edge (240). Quartile values are similar, though the upper quartile is slightly higher at nadir.

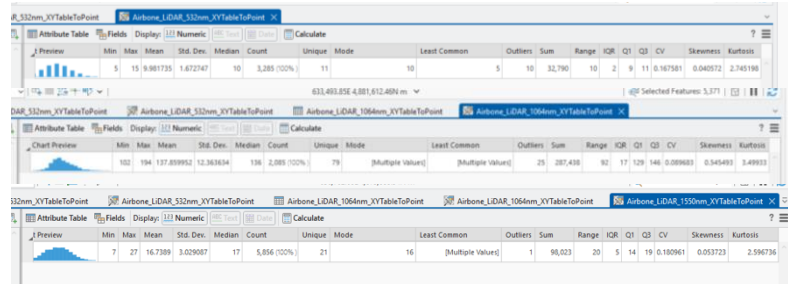
The intensity differences in the nadir region and the swath edge is likely due to scan geometry effects. At nadir, the laser beam hits the target at a smaller incident angle, usually close to 0, while at swath edge, the laser beam hits the target at larger incident angle. The backscatter is stronger with smaller incident angle, since the intensity is proportional to $\cos(\theta)$. Furthermore, the range increases at swath edge, which increases the footprint size, thus reducing the signal strength.

Therefore, even though both polygons represent the same woodland land cover type, the scan angle and range can cause intensity value variations.

Appendix

A. Screenshots and statistics of intensity values by land cover

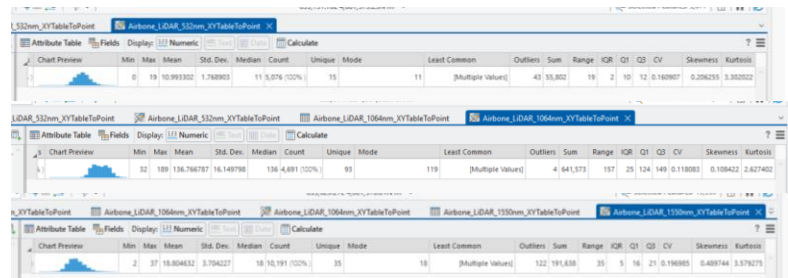
SoilCover



Intensity statistics

Wavelength	Mean	Max	Min	Q1	Q3
532nm	9.98	5	15	9	11
1064nm	137.86	194	102	129	146
1550nm	16.74	27	7	14	19

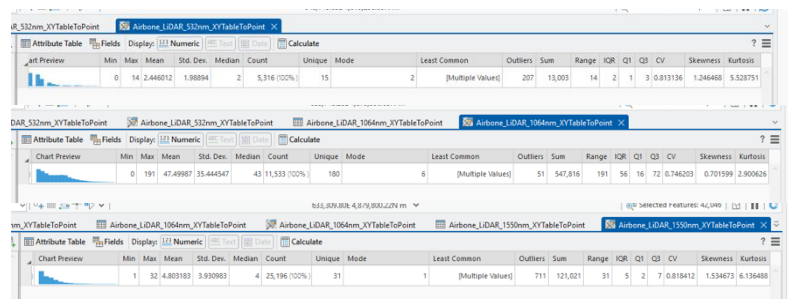
GrassCover



Intensity statistics

Wavelength	Mean	Max	Min	Q1	Q3
532nm	10.99	19	0	10	12
1064nm	136.77	189	32	124	149
1550nm	18.80	37	2	16	21

TreeCover



Intensity statistics					
Wavelength	Mean	Max	Min	Q1	Q3
532nm	2.45	14	0	1	3
1064nm	47.50	191	0	16	72
1550nm	4.80	32	1	2	7